Hamiltonian of Mean Force Beyond the Commuting Gaussian Benchmark

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We study the operator representability problem for the Hamiltonian of mean force (HMF) in Gaussian linearly coupled environments beyond the commuting limit. Building on the quenched-density construction, we derive an exact imaginary-time reformulation in which bath statistics enter as scalar kernel moments while operator growth is generated by the adjoint chain $\{\mathrm{ad}^n_{H_Q}(f)\}_{n\geq 0}$. This yields a structural criterion: a finite closed-form HMF exists only when the adjoint-generated operator algebra closes inside the target ansatz. We then instantiate the criterion in a Pauli-closed qubit model and obtain an explicit closed-form mean-force Hamiltonian with temperature- and bath-dependence carried by mode-resolved kernel coefficients. The manuscript is framed as the noncommuting Gaussian follow-up to the commuting benchmark analysis of arXiv:2602.13146 and sets up the non-Gaussian extension program.

INTRODUCTION

This manuscript is a direct follow-up to Ref. [1] (arXiv:2602.13146). That paper established the quenched-density representation and delivered an exact benchmark in the commuting Gaussian sector. The present paper starts from that benchmark and addresses the next step: how the Hamiltonian of mean force is organised once the coupling operator does *not* commute with the bare system Hamiltonian.

In closed quantum statistical mechanics, equilibrium is generated by a Hamiltonian: $\rho \propto e^{-\beta H}$. For an open system with finite coupling, the operationally defined equilibrium state of the subsystem is the reduced state of the global Gibbs ensemble,

$$\bar{\rho}_Q(\beta) = \mathrm{Tr}_X\, e^{-\beta H_{\mathrm{tot}}}, \tag{1}$$

$$e^{-\beta H_{\mathrm{MF}}(\beta)} \propto \frac{\mathrm{Tr}_X e^{-\beta H_{\mathrm{tot}}}}{Z_X(\beta)}, \quad Z_X(\beta) = \mathrm{Tr}_X e^{-\beta H_X}. \tag{2}$$

This object is generally not $e^{-\beta H_Q}$ for the bare system Hamiltonian H_Q. The resulting representational question is precise: what operator, if any, plays the role of an equilibrium generator for the subsystem once the coupling is non-negligible? The Hamiltonian of mean force (HMF) answers this by construction and is the standard starting point in strong-coupling thermodynamics and the mean-force Gibbs-state program [2–7].

Why care about a mean-force Hamiltonian at all? First, it provides the exact reduced equilibrium object that underlies strong-coupling thermodynamic identities, including free-energy and work relations that remain valid beyond weak coupling[2, 5, 8–11]. Second, it furnishes a consistent equilibrium initialization for open-system dynamics when correlations with the bath are unavoidable, a point emphasized in the literature on correlated initial states and reduced dynamics [4, 12]. Related subensemble approaches explicitly treat the subsystem thermodynamics as inherited from a global canonical ensemble, making the effective reduced generator central to the formalism[13]. In short, the HMF is not an optional reinterpretation; it is the exact operator that encodes the reduced equilibrium state whenever system–bath coupling is finite.

Historically, open-system theory prioritized weak-coupling and Markovian regimes, where reduced equilibrium can often be approximated by a Gibbs state of a renormalized H_Q. At finite coupling the reduced state inherits explicit temperature dependence and interaction-induced operator content that is not captured by a simple renormalization. Coupling-dependent thermodynamic response features in quantum Brownian motion and related models highlight this complexity[14, 15]. The strong-coupling literature consequently treats the mean-force Gibbs state as a distinct equilibrium object, with operational ramifications for heat and energy definitions[9, 16–18]. The HMF can encode interaction-induced operator content absent from H_Q [14, 15, 19], making explicit representation of this content a primary goal.

The HMF literature itself is broad but structured. Canonical definitions and thermodynamic identities are developed in strong-coupling thermodynamics and fluctuation-relation work[2, 3, 5, 8]. A comprehensive review consolidates the "static" mean-force Gibbs perspective with the "dynamical" return-to-equilibrium perspective in open quantum systems [4]. Operational questions such as measurability and thermodynamic consistency at strong coupling have also been pursued[9, 17, 18].

Exact or controlled evaluations exist in special cases. For commuting (QND) interactions, the operator algebra closes trivially and the HMF can be written explicitly[20]. Quadratic/Gaussian models (e.g., damped harmonic oscillators) are solvable because Gaussianity is preserved, leading to closed operator forms[21–23]. Finite-dimensional closures such as spin-boson or single-qubit models provide additional controlled benchmarks[24]. Beyond these cases, the mean-force Gibbs state is often accessed through systematic limits: Cresser and Anders derive weak- and ultrastrong-coupling expressions and

show that, in the ultrastrong limit, the mean-force Gibbs state becomes diagonal in the interaction basis rather than the system Hamiltonian basis[25, 26]. Recent work generalizes the HMF framework to finite baths by introducing a pair of quantum Hamiltonians of mean force that incorporate bath feedback[27], and structural studies further analyze the operator content and potential renormalization of the HMF in extended settings[19, 28].

Outside solvable models, most approaches are perturbative or numerical. Weak-coupling expansions yield controlled but limited series [25, 29]. Reaction-coordinate mappings[30–33] and polaron transforms provide approximate local generators applicable at intermediate coupling. Imaginary-time path-integral methods, stochastic representations, and hierarchical-equations techniques can compute $\bar{\rho}_Q(\beta)$ directly but typically do not yield compact operator forms for $H_{\rm MF}$[34–39]. Stochastic Liouville and partition-free approaches give complementary numerical access without yielding closed-form HMFs [40, 41]. These methods are indispensable for quantitative predictions but leave open the representational question: when does a local or otherwise restricted operator form exist?

The influence-functional formalism is a natural language for this problem. For Gaussian baths with linear coupling it provides an exact route to integrating out bath degrees of freedom[21, 22, 42]. In equilibrium it becomes a Euclidean (imaginary-time) influence functional, usually bilocal in τ, and admits a Hubbard–Stratonovich rewriting as a quenched Gaussian-field average[43–45]. Related path-integral derivations of quantum Langevin dynamics make explicit the noise and dissipation structure inherited from the bath[46, 47]. This formulation also clarifies the terminology: "nonlocal" initially refers to imaginary-time nonlocality of the kernel, whereas the HMF locality question concerns the operator structure on the system Hilbert space.

The existence of $H_{\rm MF}$ is therefore not the issue—it is defined by a logarithm of a traced exponential. The real obstruction is representability: when does $H_{\rm MF}$ admit a closed-form expression within a restricted operator family (few-body, spatially local, or a given algebra)? Much of the literature either (i) solves special models, (ii) expands in controlled limits, or (iii) computes $\bar{\rho}_Q(\beta)$ numerically and analyzes its properties, but a general structural criterion is still lacking [25, 27, 28].

Relative to Ref. [1], the present manuscript has a narrower but deeper scope. We retain the exact Gaussian influence-functional and quenched-field identities, then use adjoint-action and Magnus/BCH organisation [48–50] to study representability in the noncommuting sector. The central result is an explicit closure criterion: a finite closed-form HMF exists only when the operator algebra generated by repeated adjoint action of H_Q on the coupling operators closes inside the target ansatz. We then instantiate this criterion in a Pauli-closed qubit example, where all bath dependence is reduced to scalar kernel moments and the mean-force Hamiltonian is written in exact closed form. Moreover, it is not often one has the opportunity to say something new about a qubit.

QUENCHED REPRESENTATION AND INFLUENCE FUNCTIONAL

We begin by recapitulating the quenched representation introduced in Ref. [1]. To proceed directly from Sec. , we explicit the composite model. We denote the bare system Hamiltonian by H_Q and write

$$H_{\rm tot} = H_Q + H_X + H_{\rm int}, \tag{3}$$

where H_X is the bath Hamiltonian. We assume a factorizable interaction

$$H_{\rm int} = f \otimes B, \tag{4}$$

where f acts on the system and B is a bath operator. We can define the reduced equilibrium operator (up to normalisation) by $\bar{\rho}_Q(\beta) \equiv {\rm Tr}_X e^{-\beta H_{\rm tot}}$. As shown in Ref. [1], this can be represented as a *quenched density*. This is an average over a stochastic propagator, given by:

$$\bar{\rho}_Q(\beta) = \mathbb{E}_\xi\big[U_\xi(\beta)\big], \tag{5}$$

$$U_\xi(\beta) \equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\,\big(H_Q + \xi(\tau) f\big)\right], \tag{6}$$

where $\xi(\tau)$ is a stochastic process whose statistics encode the bath correlations, and $\mathcal{T}_\tau$ denotes time-ordering in imaginary time. Regardless of the precise form of bath and coupling, the reduced density must always be describable in this form.

This result can be understood intuitively by first observing that since the environment influence can only enter through the system coupling f, its influence can be captured by attaching a τ-dependent driving field to f [51, 52]. If this were a single deterministic field however, it would correspond to the bath exerting the *same* back-action history for every microscopic bath configuration. But ${\rm Tr}_X$ averages over many bath microstates in the thermal ensemble, and hence over many back-action histories. In this sense the partial trace is necessarily an average over histories in imaginary time, and the quenched representation simply makes this averaging explicit. For each realisation $\xi(\tau)$ the system evolves under an imaginary-time Hamiltonian $H_Q + \xi(\tau) f$; the bath is then recovered by averaging over $\xi(\tau)$ with a law chosen to reproduce the bath-induced correlations. In this sense $\xi(\tau)$ is not a physical external control field but an efficient parametrisation of the bath history $B(\tau)$ as seen through the coupling channel.

We may understand what formal properties are demanded of ξ by considering the influence functional it is

required to match. Working in imaginary time, introduce the bath interaction picture:

$$B(\tau) \equiv e^{\tau H_X} B e^{-\tau H_X}, \qquad \tau \in [0, \beta], \tag{7}$$

and define the bath thermal state $\rho_X \equiv e^{-\beta H_X}/Z_X$. For an arbitrary c-number source $j(\tau)$ coupled linearly to $B(\tau)$, the bath generates a (time-ordered) functional

$$\mathcal{Z}_X[j] \equiv \mathrm{Tr}_X \left[\mathcal{T}_\tau \exp\left(-\int_0^\beta d\tau\, j(\tau)\, B(\tau) \right) \rho_X \right]. \tag{8}$$

Here $j(\tau)$ is introduced as an external c-number source used to generate ordered bath correlators by functional differentiation. More generally, $j(\tau)$ may be any object commuting with the bath algebra (e.g. a system operator tensored with $\mathbb{I}_X$). In the influence-functional derivation it is ultimately supplied by the system history, which becomes a c-number function in the path-integral representation. In the influence-functional approach, tracing out the bath produces precisely such a functional, evaluated on the system history through the coupling channel.

From this, the bath contribution to the effective Euclidean action can be written as [21, 22, 42]

$$\mathcal{F}[f] \equiv \mathcal{Z}_X[f], \qquad \Phi[f] \equiv \log \mathcal{F}[f]. \tag{9}$$

A key structural fact is that $\Phi[f] = \log \mathcal{F}[f]$ is the *cumulant generating functional* of the bath operator $B(\tau)$ with respect to the thermal state. Concretely, the generalised (time-ordered) cumulant theorem implies the connected expansion [53, 54]

$$\begin{aligned} \Phi[f] = \sum_{n\geq 1} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \\ \times K^{(n)}(\tau_1, \ldots, \tau_n)\, f(\tau_1) \cdots f(\tau_n), \end{aligned} \tag{10}$$

where the kernels $K^{(n)}$ are the *connected* (cumulant) bath correlators, defined via [55]

$$K^{(n)}(\tau_1, \ldots, \tau_n) \equiv \left\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \right\rangle_c. \tag{11}$$

The explicit connection back to Eq. (8) is given via its functional differentiation:

$$\left\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \right\rangle_c = (-1)^n \left. \frac{\delta^n \log \mathcal{Z}_X[j]}{\delta j(\tau_1) \cdots \delta j(\tau_n)} \right|_{j=0}. \tag{12}$$

The bath influence is then completely characterised by the hierarchy $\{K^{(n)}\}_{n\geq 1}$.

To connect the influence functional back to the quenched density, we use the fact that the bath influence depends on the system history only through the linear functional $\int_0^\beta d\tau\, f(\tau) B(\tau)$. One may therefore represent $\mathcal{F}[f]$ as the (generalised) characteristic functional of an auxiliary field $\xi(\tau)$ [43–45]:

$$\mathcal{F}[f] = \mathbb{E}_\xi \left[\exp\left(-\int_0^\beta d\tau\, \xi(\tau)\, f(\tau) \right) \right], \tag{13}$$

where $\mathbb{E}_\xi[\cdot]$ denotes averaging with respect to a (possibly complex) measure on ξ-histories chosen such that Eq. (13) holds.

Since $\xi(\tau)$ is a commuting c-number field, its n-point moments are symmetric under permutations of the time arguments. The influence kernels $K^{(n)}$ fix the *cumulants* of ξ via

$$\left\langle \xi(\tau_1) \cdots \xi(\tau_n) \right\rangle_c = K^{(n)}(\tau_1, \ldots, \tau_n). \tag{14}$$

Consequently, the ordinary correlation functions of the noise are obtained from $\{K^{(m)}\}$ by the standard moment-cumulant relations:

$$\left\langle \xi(\tau_1) \cdots \xi(\tau_n) \right\rangle = \sum_{\pi \in \mathcal{P}_n} \prod_{C \in \pi} K^{(|C|)}\big(\{\tau_i\}_{i\in C}\big), \tag{15}$$

where $\mathcal{P}_n$ denotes the set of all partitions of the index set $\{1, \ldots, n\}$, and C are the disjoint blocks of a given partition $\pi \in \mathcal{P}_n$. After shifting the mean so that $K^{(1)}(\tau) = \langle \xi(\tau) \rangle = 0$, one has (for example)

$$\left\langle \xi(\tau_1)\xi(\tau_2) \right\rangle = K^{(2)}(\tau_1, \tau_2), \tag{16}$$

$$\left\langle \xi(\tau_1)\xi(\tau_2)\xi(\tau_3) \right\rangle = K^{(3)}(\tau_1, \tau_2, \tau_3), \tag{17}$$

$$\begin{aligned} &\left\langle \xi(\tau_1)\xi(\tau_2)\xi(\tau_3)\xi(\tau_4) \right\rangle \\ &\quad = K^{(4)}(\tau_1, \tau_2, \tau_3, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_2) K^{(2)}(\tau_3, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_3) K^{(2)}(\tau_2, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_4) K^{(2)}(\tau_2, \tau_3). \end{aligned} \tag{18}$$

QUENCHED DENSITY AND THE HAMILTONIAN OF MEAN FORCE

The Hamiltonian of mean force is, by definition, the operator whose Gibbs form reproduces the system's reduced equilibrium state. Equivalently, it is the *operator logarithm* of the unnormalised reduced equilibrium operator

$$\bar{\rho}_Q(\beta) \equiv \mathrm{Tr}_X e^{-\beta H_{\mathrm{tot}}}, \qquad H_{\mathrm{MF}}(\beta) \;\equiv\; -\frac{1}{\beta} \log \bar{\rho}_Q(\beta), \tag{19}$$

defined up to an additive multiple of the identity (fixed only when normalising the state).

The quenched representation in Eq. (5) supplies an exact stochastic parametrisation of the unnormalised mean-force Gibbs operator $\bar{\rho}_Q(\beta)$ by making the bath trace an explicit average over imaginary-time back-action histories. Combining Eq. (5) with Eq. (19) yields

$$H_{\mathrm{MF}}(\beta) = -\frac{1}{\beta} \log \mathbb{E}_\xi[U_\xi(\beta)]\,. \tag{20}$$

Thus, constructing the mean-force Hamiltonian reduces to evaluating a stochastic average and then compressing

the result via an operator logarithm. The conditions for being able to perform this average exactly is the focus of the present work.

To make the handling of this problem more concrete, we shall specialise the environment to the Caldeira-Leggett model, where the bath is a collection of harmonic oscillators ($H_X = \sum_k \omega_k b_k^\dagger b_k$) and the coupling is linear in bath coordinates ($B = \sum_k c_k x_k$). None of the results that follow are essentially dependent on this choice, and a generalisation to anharmonic environments is (relatively) straightforward. In the interests of comprehensibility however, we restrict our scope to quadratic environments. In this setting, the auxiliary field $\xi(\tau)$ becomes a stationary zero-mean Gaussian process completely characterized by its covariance

$$\mathbb{E}_\xi[\xi(\tau)\xi(\tau')] = K(\tau - \tau'). \tag{21}$$

The kernel $K(\tau)$ is determined by the bath spectral density $J(\omega) = \frac{\pi}{2}\sum_k \frac{c_k^2}{m_k\omega_k}\delta(\omega - \omega_k)$ via the relation [56]

$$K(\tau) = \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\,\frac{\cosh\big(\omega(\beta/2 - |\tau|)\big)}{\sinh(\beta\omega/2)}. \tag{22}$$

A key property of this kernel is its integrated strength. Integrating Eq. (22) yields

$$\begin{aligned}\int_0^\beta d\tau\, K(\tau) &= \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\int_0^\beta d\tau\,\frac{\cosh\big(\omega(\beta/2 - |\tau|)\big)}{\sinh(\beta\omega/2)}\\ &= \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\,\frac{2}{\omega}\\ &= 2\lambda,\end{aligned} \tag{23}$$

where λ is the explicit reorganisation energy. Consequently, the total variance of the integrated noise field $\Xi = \int_0^\beta d\tau\,\xi(\tau)$ grows linearly with inverse temperature:

$$\mathbb{E}_\xi[\Xi^2] = \int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau - \tau') = 2\beta\lambda. \tag{24}$$

In the case that the system and its coupling commute, $[H_Q, f] = 0$ and f is τ-independent in imaginary time. In this instance time ordering drops out, and the average is given by [1]:

$$\bar{\rho}_Q(\beta) = \exp\left[-\beta\left(H_Q - \frac{\kappa_0(\beta)}{2}f^2\right)\right], \tag{25}$$

which in turn yields

$$H_{\rm MF}(\beta) = H_Q - \frac{\kappa_0(\beta)}{2}f^2 + \frac{1}{\beta}\log Z_X(\beta)\,\mathbb{I}. \tag{26}$$

Notably this correction to $H_{\rm MF}(\beta)$ is entirely *classical* [1]. This is hardly surprising, but emphasises that truly quantum effects *always* stem from non-commutativity. Truly quantum thermodynamic effects are therefore only present when $[H_Q, f] \neq 0$. In this case however, the noise enters through a noncommuting operator inside $\mathcal{T}_\tau$, rendering the question of averaging highly non-trivial. In the next section, we attack this problem directly, deriving conditions under which $H_{\rm MF}(\beta)$ possesses a closed form.

CLOSURE OF THE HAMILTONIAN OF MEAN FORCE

The obstruction to writing $H_{\rm MF}(\beta)$ in a compact operator form is not the existence of the mean-force object (it is defined by a logarithm), but the *representability* of that logarithm inside a restricted operator family (few-body, local, Pauli strings, etc.). In the harmonic case this representability question reduces to a precise closure problem. To show this, we first write the quenched propagator in the imaginary-time interaction picture with respect to H_Q:

$$\begin{aligned}U_\xi(\beta) &= e^{-\beta H_Q}W_\xi(\beta),\\ W_\xi(\beta) &\equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\,\xi(\tau)\,\tilde{f}(\tau)\right],\end{aligned} \tag{27}$$

where $\tilde{f}(\tau) \equiv e^{\tau H_Q} f e^{-\tau H_Q}$.

Because the noise is classical Gaussian with zero mean, the average of the time-ordered exponential resums exactly in terms of the second cumulant. Time ordering adds a non-trivial complication, which is addressed in Appendix A. The result may be succinctly expressed in terms of an *influence operator*

$$\begin{aligned}\langle W_\xi(\beta)\rangle_\xi &\equiv \bar{W}(\beta) = \exp\big(\Delta(\beta)\big),\\ \bar{\rho}_Q(\beta) &= \langle U_\xi(\beta)\rangle_\xi = e^{-\beta H_Q}\,e^{\Delta(\beta)},\end{aligned} \tag{28}$$

where the influence exponent $\Delta(\beta)$ is defined in accordance with Eq.Eq. (A12). It is convenient to perform a final massaging of this exponent to remove the explicit time ordering, the details of which we relegate to Appendix A. After this manipulation, $\Delta(\beta)$ takes the form

$$\Delta(\beta) \equiv \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau - \tau')\,\tilde{f}(\tau)\tilde{f}(\tau'). \tag{29}$$

To progress, we introduce the adjoint chain generated by H_Q acting on the coupling,

$$f_n \equiv \mathrm{ad}^n_{H_Q}(f), \qquad \mathrm{ad}_{H_Q}(A) \equiv [H_Q, A], \tag{30}$$

so that the interaction-picture operator has the exact series

$$\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q} = \sum_{n\geq 0}\frac{\tau^n}{n!}\,f_n. \tag{31}$$

Substituting Eq. (31) into Eq. (29) yields

$$\Delta(\beta) = \sum_{n,m\geq 0} C^{>}_{nm}(\beta) f_n f_m, \tag{32}$$

where we have further defined an *ordered kernel-moment matrix*

$$C^{>}_{nm}(\beta) \equiv \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau-\tau')\, \frac{\tau^n}{n!}\, \frac{\tau'^m}{m!}. \tag{33}$$

Here the notation $>$ is used to track the triangular domain of integration, where τ (associated with n) is the upper limit of integration for τ' (associated with m).

With this definition, all bath/temperature dependence enters $\Delta(\beta)$ only through the scalar moments $C^{>}_{nm}(\beta)$, while all operator structure is carried by the adjoint chain $\{f_n\}$. The final step is to re-express the averaged propagator in a *single* exponential. Using Eq. (28) we write

$$\bar{\rho}_Q(\beta) = e^{-\beta H_Q}\, e^{\Delta(\beta)} \equiv e^{-\beta H_{\rm MF}(\beta)}, \tag{34}$$

where the mean-force Hamiltonian satisfies

$$-\beta H_{\rm MF}(\beta) = \log\Big(e^{-\beta H_Q} e^{\Delta(\beta)}\Big). \tag{35}$$

Let $A \equiv -\beta H_Q$ and $B \equiv \Delta(\beta)$. The Baker–Campbell–Hausdorff (BCH) series [48, 49, 57] gives

$$\begin{aligned}\log(e^A e^B) = A + B + \frac{1}{2}[A,B] \\ + \frac{1}{12}[A,[A,B]] + \frac{1}{12}[B,[B,A]] + \cdots\end{aligned} \tag{36}$$

Dividing by $-\beta$ and using $A = -\beta H_Q$ yields the expansion

$$\begin{aligned}H_{\rm MF}(\beta) = H_Q - \frac{1}{\beta}\Delta(\beta) + \frac{1}{2}[H_Q, \Delta(\beta)] \\ - \frac{\beta}{12}[H_Q,[H_Q,\Delta(\beta)]] \\ + \frac{1}{12}[\Delta(\beta),[\Delta(\beta),H_Q]] + \cdots .\end{aligned} \tag{37}$$

Expressed in this way, the closure content of $H_{\rm MF}(\beta)$ is transparent: all terms are built from repeated commutators of H_Q acting on Δ, together with higher BCH terms involving commutators among those objects.

To see the explicit generator, introduce the adjoint superoperator $\mathrm{ad}_{H_Q}(\cdot) \equiv [H_Q, \cdot]$. Then Eq. (37) becomes

$$\begin{aligned}H_{\rm MF}(\beta) = H_Q - \frac{1}{\beta}\Delta + \frac{1}{2}\,\mathrm{ad}_{H_Q}\Delta \\ - \frac{\beta}{12}\,\mathrm{ad}^2_{H_Q}\Delta + \frac{1}{12}\,[\Delta, \mathrm{ad}_{H_Q}\Delta] + \cdots .\end{aligned} \tag{38}$$

Finally, because $\Delta(\beta)$ is quadratic in the adjoint chain, each commutator with H_Q simply shifts indices:

$$\mathrm{ad}_{H_Q}(f_n f_m) = [H_Q, f_n f_m] = f_{n+1} f_m + f_n f_{m+1}. \tag{39}$$

Hence the first commutator term takes the explicit form

$$[H_Q, \Delta(\beta)] = \sum_{n,m\geq 0} C^{>}_{nm}(\beta)\,\big(f_{n+1} f_m + f_n f_{m+1}\big), \tag{40}$$

and higher $\mathrm{ad}^k_{H_Q}\Delta$ generate the same family of products $f_a f_b$ with shifted indices. The remaining BCH terms (e.g. $[\Delta, \mathrm{ad}_{H_Q}\Delta]$) introduce commutators among these quadratic elements, and are the sole source of additional operator growth beyond the quadratic span. The only remaining question is whether the operator family generated by Eq. (38) can be brought to a closed form.

The preceding identities imply that the entire BCH commutator tower *linear* in Δ remains within the quadratic span $\mathrm{span}\{f_n f_m\}$. Terms to this order explicitly admit a closed resummation. Let $A \equiv -\beta H_Q$ and $B \equiv \Delta(\beta)$. Then the BCH logarithm satisfies the standard linear-in-B identity [50, 58]

$$\begin{aligned}\log\big(e^A e^B\big) = A + \Phi\big(\mathrm{ad}_A\big)\, B + O(B^2), \\ \Phi(x) \equiv \frac{x}{1-e^{-x}}.\end{aligned} \tag{41}$$

Substituting $A = -\beta H_Q$ and dividing by $-\beta$ gives the mean-force Hamiltonian to first order in Δ:

$$H_{\rm MF}(\beta) = H_Q - \frac{1}{\beta}\,\Phi\big(\beta\,\mathrm{ad}_{H_Q}\big)\,\Delta(\beta) + O(\Delta^2). \tag{42}$$

Expanding Φ yields the explicit commutator series with Bernoulli numbers B_k,

$$\begin{aligned}H_{\rm MF}(\beta) = H_Q - \frac{1}{\beta}\sum_{k\geq 0}\frac{B_k}{k!}(-\beta)^k\,\mathrm{ad}^k_{H_Q}\Delta(\beta) + O(\Delta^2), \\ \Phi(x) = \sum_{k\geq 0}\frac{B_k}{k!}(-x)^k.\end{aligned} \tag{43}$$

Using Eq. (39) repeatedly, the iterated commutators admit the closed binomial form

$$\mathrm{ad}^k_{H_Q}(f_n f_m) = \sum_{j=0}^{k}\binom{k}{j} f_{n+j}\, f_{m+k-j}, \tag{44}$$

so every term in Eq. (43) is explicitly a linear combination of products $f_a f_b$ with scalar coefficients determined solely by $C^{>}_{nm}(\beta)$ and universal combinatorics. Because of this, we may bring the first order contribution into a particularly compact form by introducing the discrete shift operator D. This will act on the moment matrix $C^{>}$ as

$$\begin{aligned}(\mathsf{D}C^{>})_{ab} \equiv C^{>}_{a-1,b} + C^{>}_{a,b-1}, \\ C^{>}_{rs} \equiv 0 \quad \text{if any index is negative.}\end{aligned} \tag{45}$$

Then Eq. (39) implies, after reindexing,

$$\mathrm{ad}_{H_Q}\Delta(\beta) = \sum_{a,b\geq 0}(\mathsf{D}C^{>}(\beta))_{ab}\, f_a f_b, \tag{46}$$

and by iteration,

$$\mathrm{ad}^k_{H_Q}\Delta(\beta) = \sum_{a,b\geq 0}(\mathsf{D}^k C^{>}(\beta))_{ab}\, f_a f_b. \tag{47}$$

Equivalently, $(\mathsf{D}^k C^>)_{ab} = \sum_{j=0}^{k} \binom{k}{j} C^>_{a-j,\, b-(k-j)}$, which is the binomial index-shift structure implied by Eq. (44). From this we obtain the following form for $H_{\rm MF}(\beta)$ to linear order in Δ:

$$H_{\rm MF}(\beta) = H_Q - \frac{1}{\beta} \sum_{a,b\geq 0} \widetilde{C}^>_{ab}(\beta)\, f_a f_b + O(\Delta^2), \tag{48}$$

where the *renormalised moment matrix* $\widetilde{C}^>(\beta)$ is the universal transform

$$\widetilde{C}^>(\beta) \equiv \Phi\big(\beta\, \mathsf{D}\big)\, C^>(\beta) = \sum_{k\geq 0} \frac{B_k}{k!} (-\beta)^k\, \mathsf{D}^k C^>(\beta). \tag{49}$$

Thus, to linear order in the influence operator, the mean-force Hamiltonian is *exactly* a quadratic form in the products $f_a f_b$, with all bath/temperature dependence entering only through the scalar matrix $\widetilde{C}^>_{ab}(\beta)$. Any operator growth beyond the quadratic span is confined to the nonlinear BCH sector $O(\Delta^2)$ (e.g. $[\Delta, \mathrm{ad}_{H_Q}\Delta]$). While these terms are more difficult to handle as they involve commutators of quadratic forms, they nevertheless admit a precise series expansion via the full BCH formula Eq. (36). With this form in hand, we can now state a theorem for the representability of $H_{\rm MF}(\beta)$ in a given operator family $\mathcal{A}$ as a *closure criterion*.

Theorem: Finite representability of mean force

Fix a target operator family $\mathcal{A}$ (e.g. k-local Pauli strings, a Lie algebra plus identity, or a finite-dimensional associative operator algebra) in which we seek to represent $H_{\rm MF}(\beta)$. The Gaussian construction above allows us to formulate the exact criteria for representability as a closure theorem. In a physical sense, these conditions determine whether a valid *local generator* can be defined for the system at all; failure of closure implies that the effective Hamiltonian is fundamentally non-local with respect to the algebra $\mathcal{A}$.

Theorem (Closure of the Mean Force). *Let $\mathcal{A}$ be a target operator space containing the bare system Hamiltonian H_Q and coupling f. The Hamiltonian of Mean Force $H_{\rm MF}(\beta)$ admits a finite representation within $\mathcal{A}$ if and only if the following three closure conditions are satisfied:*

(C1) Adjoint Closure (No linear operator growth). The subspace generated by repeated commutators of the coupling f with H_Q must remain inside $\mathcal{A}$:

$$f \in \mathcal{A}, \qquad \mathrm{ad}_{H_Q}(\mathcal{A}) \subseteq \mathcal{A}, \quad \text{equivalently} \quad \mathrm{span}\{f_n\}_{n\geq 0} \subseteq \mathcal{A}. \tag{50}$$

(C2) Quadratic Closure (No growth in the Gaussian sector). Because the exact influence operator $\Delta(\beta)$ is quadratic in the adjoint chain, the quadratic span generated by $\{f_n\}$ must also lie in $\mathcal{A}$:

$$f_n f_m \in \mathcal{A} \quad \text{for all indices that contribute in } \Delta(\beta), \tag{51}$$

so that $\Delta(\beta) \in \mathcal{A}$. A sufficient (and common) condition is that $\mathcal{A}$ is an associative algebra containing the identity and closed under multiplication.

(C3) BCH Closure (No growth from commutators among quadratic elements). To ensure that the *full* BCH logarithm $\log(e^{-\beta H_Q} e^{\Delta})$ remains in $\mathcal{A}$, the commutators generated among the quadratic elements must also close in $\mathcal{A}$; in particular

$$[\Delta(\beta),\, \mathrm{ad}^k_{H_Q}\Delta(\beta)] \in \mathcal{A} \qquad \text{for all } k \geq 0, \tag{52}$$

and similarly for the higher nested commutators appearing in the BCH series. A sufficient (and common) condition is again that $\mathcal{A}$ is a finite-dimensional associative algebra (or a Lie algebra containing Δ and closed under commutators), in which case all BCH commutators remain in $\mathcal{A}$ by construction.

Consequence (closed-form mean-force Hamiltonian). If (C1)–(C3) hold, then $\Delta(\beta) \in \mathcal{A}$ and all BCH commutators generated by Eq. (53) remain in $\mathcal{A}$. Since $H_Q \in \mathcal{A}$ by assumption, it follows that

$$\bar{\rho}_Q(\beta) = e^{-\beta H_{\rm MF}(\beta)}, \qquad -\beta H_{\rm MF}(\beta) = \log\Big(e^{-\beta H_Q} e^{\Delta(\beta)}\Big), \tag{53}$$

defines an $H_{\rm MF}(\beta) \in \mathcal{A}$ and hence a closed-form representation of $H_{\rm MF}(\beta)$ (up to the usual additive scalar fixed by Z_X). If either (C1) fails (adjoint-chain growth) or (C2)–(C3) fail (growth in the quadratic/BCH sector), then $\Delta(\beta)$ and/or the BCH commutator tower generates operators outside $\mathcal{A}$; any representation within $\mathcal{A}$ is necessarily truncated or projected.

This statement leads to an important corollary. For continuous variable systems $H_Q = p^2/2m + V(x)$, if the potential $V(x)$ is a polynomial of degree greater than 2, the adjoint chain $\{f_n\}$ generally does not close in any finite-dimensional Lie algebra; instead, it generates the full universal enveloping algebra. Consequently, no finite representation of the mean force exists within the original operator class. This obstruction is analogous (in spirit) to the Groenewold–Van Hove no-go theorem on consistent quantisation of polynomial observables: the failure to close the Lie algebra under adjoint action corresponds to the impossibility of mapping the quantum evolution of non-quadratic potentials to a finite phase space flow. For such potentials, there is no *local operator* that exactly describes the reduced equilibrium state.

From a practical standpoint, the closure of $\mathcal{A}$ is not necessary for an arbitrarily accurate approximation of $H_{\rm MF}(\beta)$, as the BCH logarithm $\log(e^{-\beta H_Q} e^{\Delta})$ still yields a controlled expansion. The operator content is generated by the adjoint chain and its quadratic products, while all bath dependence remains scalar through

$C^{>}_{nm}(\beta)$. One may therefore truncate (in commutator depth, locality class, or operator weight) without introducing any approximation on the bath side.

This algebraic perspective offers a unifying view of strong-coupling approximations. Numerical schemes such as the polaron transformation [30, 56], hierarchical equations of motion [37], and chain-mapping/DMRG techniques [39] can be understood as distinct choices of the target operator family $\mathcal{A}$. The polaron frame corresponds to dressing $\mathcal{A}$ with coherent displacements; HEOM truncates the memory depth of the influence functional (dual to the operator degree in the adjoint chain), and chain mappings truncate the spatial extent of the harmonic bath. The 'closure' problem is thus equivalent to identifying a low-dimensional subalgebra that approximately contains the logarithm of the quenched propagator.

EXACT SOLUTION OF THE SPIN-BOSON MODEL: A RESPONSE-KERNEL ROUTE

We now put the machinery of the previous section to work. The transverse-coupling spin-boson model is the ideal proving ground: the $\mathfrak{su}(2)$ algebra is just rich enough to generate non-trivial dynamics while remaining analytically tractable at every coupling strength. Moreover, it is not often one has the opportunity to say something new about a qubit.

There are a number of approaches to performing this calculation. Most directly, one could follow the chain of closure conditions $(C1)-(C3)$ of Sec. , calculating first f_n, then the products $f_n f_m$, then finally the full chain of BCH commutators. This is a thoroughly miserable exercise in algebraic manipulation, but for the truly masochistic reader we provide details of it in App. B.

Rather than torturing ourselves with commutator expansions, we shall adopt a more direct approach, which is not only less algebraically intensive but leads to a more robust physical interpretation of the result. We bypass the cumulant series entirely by working in the ladder basis from the outset, evaluating the product $\tilde{f}(\tau)\tilde{f}(\tau')$ in Eq. (29) directly. This naturally implies the existence of ordered response kernels, which emerge as the fundamental constituents of the solution. The same physics emerges, but the dependence on the underlying properties of the bath is more transparent.

Following Refs. [25, 26], we take

$$H_Q = \frac{\omega_q}{2}\sigma_z, \qquad f = \cos\theta\,\sigma_z - \sin\theta\,\sigma_x, \tag{54}$$

and write $c \equiv \cos\theta$, $s \equiv \sin\theta$ throughout. The coupling operator mixes a *commuting* sector, $c\sigma_z$ (which is dark to the system dynamics), with a *transverse* sector $-s\sigma_x$ that precesses under H_Q.

Rather than expanding in the cumulant series, we compute the imaginary-time interaction picture operator directly. Introducing the standard raising and lowering operators $\sigma_\pm = (\sigma_x \pm i\sigma_y)/2$ and using the Heisenberg-picture identity $e^{\tau H_Q}\sigma_\pm e^{-\tau H_Q} = e^{\pm\omega_q\tau}\sigma_\pm$, the coupling operator in the interaction picture is

$$\tilde{f}(\tau) = c\,\sigma_z - s\big(e^{\omega_q\tau}\sigma_+ + e^{-\omega_q\tau}\sigma_-\big)\,. \tag{55}$$

The ladder representation is tailor-made for computing products, since the $\sigma_\pm$ algebra is particularly transparent: $\sigma_+\sigma_- = (\mathbb{I}+\sigma_z)/2$, $\sigma_-\sigma_+ = (\mathbb{I}-\sigma_z)/2$, and $\sigma_\pm^2 = 0$. These elementary identities, applied to the product $\tilde{f}(\tau)\tilde{f}(\tau')$, immediately give

$$\begin{aligned}\tilde{f}(\tau)\tilde{f}(\tau') = &\Big[c^2 + s^2\cosh\big(\omega_q(\tau-\tau')\big)\Big]\mathbb{I}\\ &+ s^2\sinh\big(\omega_q(\tau-\tau')\big)\sigma_z\\ &+ cs\Big(e^{\omega_q\tau} - e^{\omega_q\tau'}\Big)\,\sigma_+\\ &+ cs\Big(e^{-\omega_q\tau'} - e^{-\omega_q\tau}\Big)\,\sigma_-.\end{aligned} \tag{56}$$

This deceptively compact expression is already the crux of the calculation. Each Pauli channel carries a distinct frequency dependence: the longitudinal channel σ_z is driven by the *difference* $\tau-\tau'$ alone (a function of the time-separation, not the individual times), while the transverse channels $\sigma_\pm$ are sensitive to each time argument individually. Physically, this reflects the fact that σ_z commutes with H_Q and so cannot exchange quanta with the bath, whereas $\sigma_\pm$ do not commute and can.

The ordered Green's kernel

Given the product Eq. (56), we can now evaluate the folded influence operator Eq. (29) in the ladder basis. Before doing so, it is useful to define a new object, as the-ordered integrals that appear in each channel of Eq. (58) share a common structure. They are all instances of an *ordered Green's kernel*

$$\mathcal{G}^{>}(x,y) = \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau-\tau')\, e^{x\tau + y\tau'}, \tag{57}$$

Substituting Eq. (56) into the folded influence operator Eq. (29) immediately yields

$$\Delta(\beta) = \Delta_0(\beta)\,\mathbb{I} + \Delta_z(\beta)\,\sigma_z + \Sigma_+(\beta)\,\sigma_+ + \Sigma_-(\beta)\,\sigma_-. \tag{58}$$

with channel amplitudes

$$\Sigma_+ = cs\big[\mathcal{G}^{>}(\omega_q,0) - \mathcal{G}^{>}(0,\omega_q)\big]\,, \tag{59}$$

$$\Sigma_- = cs\big[\mathcal{G}^{>}(0,-\omega_q) - \mathcal{G}^{>}(-\omega_q,0)\big]\,, \tag{60}$$

$$\Delta_z = \frac{s^2}{2}\big[\mathcal{G}^{>}(\omega_q,-\omega_q) - \mathcal{G}^{>}(-\omega_q,\omega_q)\big]\,, \tag{61}$$

$$\Delta_0 = c^2\mathcal{G}^{>}(0,0) + \frac{s^2}{2}\big[\mathcal{G}^{>}(\omega_q,-\omega_q) + \mathcal{G}^{>}(-\omega_q,\omega_q)\big]\,. \tag{62}$$

It is worth now lingering a moment on $\mathcal{G}^{>}(x,y)$. It not only allows for a compact representation of the influence operator, but possesses rich formal analogues that deepen its physical interpretation. Structurally, it is the imaginary-time analogue of the *two-time retarded Green's function* of the bath, computed on the triangular domain $0 \le \tau' \le \tau \le \beta$ and projected onto the Laplace frequencies (x,y) set by the system's own eigenoperators. Furthermore, this object provides a direct bridge to the Matsubara formalism. If one identifies (x,y) with analytic continuations of discrete frequencies $i\omega_n$, $\mathcal{G}^{>}$ captures the exact bath response summed over all thermal modes. The specific values $x,y \in \{0,\pm\omega_q\}$ selected by the algebra thus correspond to evaluating this effective vertex on the physical mass shell of the system.

The "greater" superscript is deliberate: $\mathcal{G}^{>}$ keeps $\tau > \tau'$ (time-ordered), in precise analogy with the real-time Greater Green's function $G^{>}(t,t') = \langle A(t)B(t')\rangle$ familiar from non-equilibrium field theory. Moreover, it acts as a *generating function* for the ordered kernel moments $C^{>}_{nm}$ of Sec. . This is made transparent by the identification

$$\mathcal{G}^{>}(x,y) = \sum_{n,m} C^{>}_{nm} x^n y^m \tag{63}$$

such that the derivatives at the origin recover moments in the usual generating function fashion. The one definitional difference is that the moments are scaled by their combinatorial weights, i.e.

$$\partial_x^n \partial_y^m \; \mathcal{G}^{>}(x,y)\big|_{x,y=0} = n!m!\, C^{>}_{nm}. \tag{64}$$

Having defined $\mathcal{G}^{>}$, the existence of an anti-time-ordered partner $\mathcal{G}^{<}$ over the complementary domain $\tau' > \tau$ is immediately implied:

$$\mathcal{G}^{<}(x,y) = \int_0^\beta d\tau \int_\tau^\beta d\tau'\, K(\tau-\tau')\, e^{x\tau + y\tau'}. \tag{65}$$

The Kubo-Martin-Schwinger (KMS) symmetry $K(-u) = K(u)$ then immediately implies the crossing relation

$$\mathcal{G}^{<}(x,y) = \mathcal{G}^{>}(y,x). \tag{66}$$

. The KMS condition further implies a fluctuation theorem for $\mathcal{G}^{>}$ itself,

$$\mathcal{G}^{>}(x,y) = e^{\beta(x+y)}\, \mathcal{G}^{>}(-y,-x), \tag{67}$$

which is the imaginary-time echo of the KMS relation and links emission to absorption with the Boltzmann factor $e^{\beta(x+y)}$.

This generating-function perspective pays immediate dividends. Changing variables to $u = \tau - \tau'$ and $v = \tau'$ and using Eq. (69), the double integral collapses to the one-dimensional Laplace transform,

$$\mathcal{G}^{>}(x,y) = \frac{e^{x\beta}\mathcal{K}(y) - \mathcal{K}(x)}{x+y}, \tag{68}$$

where we have introduced the *bare Laplace transform* of the bath kernel,

$$\mathcal{K}(\omega) \equiv \int_0^\beta du\, K(u)\, e^{\omega u}. \tag{69}$$

The closed form Eq. (68) is one of the central results of this framework: the entire bath response relevant to the qubit, at *all* coupling strengths, is encoded in the function $\mathcal{K}(\omega)$ evaluated at just three frequencies, $\omega \in \{0,\pm\omega_q\}$. This is not a weak-coupling approximation; it is an exact consequence of the $\mathfrak{su}(2)$ closure and persists at arbitrary coupling.

Expressions of this type have appeared in related contexts. In the Feynman-Vernon path-integral formulation [42], the influence phase $\Phi[q,q']$ involves double time-integrals of the form $\iint d\tau\, d\tau'\, \eta(\tau)\, K(\tau-\tau')\, \xi(\tau')$ over the forward and backward paths η, ξ; $\mathcal{G}^{>}(x,y)$ is precisely the generating function of these influence-phase matrix elements when the paths are restricted to those of a two-level system. In the language of diagrammatic perturbation theory, $\mathcal{K}(\omega)$ plays the role of the bath *self-energy* $\Sigma(\omega)$ on the imaginary frequency axis - a single dressed propagator that encodes the renormalisation of the qubit by its environment.

The one case that demands special treatment is the *resonant channel* $x+y=0$, where Eq. (68) becomes $0/0$. Taking the limit carefully gives

$$\mathcal{R}(\omega_q) \equiv \mathcal{G}^{>}(\omega_q,-\omega_q) = \int_0^\beta du\, (\beta-u)\, K(u)\, e^{\omega_q u}. \tag{70}$$

This has a pleasing physical interpretation, due to the factor $(\beta - u)$ in Eq.Eq. (70). This is the "available imaginary time" after the bath correlation at separation u has occurred, so the resonant kernel accumulates bath memory over the entire thermal interval, weighted by how much time remains. In the real-time formulation, the analogous resonant contribution would diverge (secular growth), but here it is regulated by the finitude of temperature.

From the reflection symmetry $K(\beta-u) = K(u)$, the Laplace transform satisfies the detailed-balance identity

$$\mathcal{K}(-\omega) = e^{-\beta\omega}\mathcal{K}(\omega). \tag{71}$$

This is, in effect, a statement of the fluctuation-dissipation theorem (FDT) in Laplace space: positive and negative frequencies are related by a Boltzmann weight, as one expects for a thermal bath. Defining

$$\mathcal{K}^{\pm} \equiv \frac{1}{2}\big(\mathcal{K}(\omega_q) \pm \mathcal{K}(-\omega_q)\big), \tag{72}$$

the even ($\mathcal{K}^{+}$) and odd ($\mathcal{K}^{-}$) combinations satisfy

$$\frac{\mathcal{K}^{+}}{\mathcal{K}^{-}} = \frac{1+e^{-\beta\omega_q}}{1-e^{-\beta\omega_q}} = \coth\!\left(\frac{\beta\omega_q}{2}\right), \tag{73}$$

precisely the Einstein fluctuation-dissipation kernel. In the spectral representation $\mathcal{K}(\omega_q) = \int d\nu\, J(\nu)\,[\dots]$, one recognises $\mathcal{K}^-$ as proportional to the *imaginary part* of the bath Green's function (dissipation), and $\mathcal{K}^+$ as proportional to its *real part* (dispersion), with Eq. (73) playing the role of the Kramers-Kronig partner.

Substituting Eq. (68) into Eq. (59) and Eq. (60) and simplifying with KMS gives

$$\Sigma_+ = \frac{cs}{\omega_q}\left[\left(1+e^{\beta\omega_q}\right)\mathcal{K}(0) - 2\mathcal{K}(\omega_q)\right], \tag{74}$$

$$\Sigma_- = \frac{cs}{\omega_q}\left[\left(1+e^{-\beta\omega_q}\right)\mathcal{K}(0) - 2\mathcal{K}(-\omega_q)\right], \tag{75}$$

from which the KMS relation between the two off-diagonal amplitudes follows immediately:

$$\Sigma_+ = e^{\beta\omega_q}\Sigma_-. \tag{76}$$

This is the propagation of the earlier detailed-balance condition Eq. (73), with the ratio of the amplitude for raising to lowering set purely by the Boltzmann factor $e^{\beta\omega_q}$, irrespective of the detailed bath spectrum. Translating to the Cartesian Bloch basis via $\Sigma_\pm = \Delta_x \pm i\Delta_y$, the final channel amplitudes are

$$\Delta_x(\beta) = \frac{cs}{\omega_q}\left[\left(1+\cosh(\beta\omega_q)\right)\mathcal{K}(0) - 2\mathcal{K}^+\right], \tag{77}$$

$$\Delta_y(\beta) = i\frac{cs}{\omega_q}\left[\sinh(\beta\omega_q)\mathcal{K}(0) - 2\mathcal{K}^-\right], \tag{78}$$

$$\Delta_z(\beta) = s^2\mathcal{R}^-, \tag{79}$$

where in analogy with $\mathcal{K}^-$, $\mathcal{R}^-$ is the antisymmetric resonant kernel response:

$$\mathcal{R}^- = \frac{1}{2}(\mathcal{R}(\omega_q) - \mathcal{R}(-\omega_q)). \tag{80}$$

The architecture of these expressions reveals the physics clearly. The transverse channels are controlled by $\mathcal{K}(0)$ and $\mathcal{K}(\omega_q)$ - the bath kernel sampled at zero frequency and at the system's natural frequency - while the longitudinal channel Δ_z is set by the *resonant* kernel $\mathcal{R}$. The separation is not accidental: it reflects the distinct sensitivity of commuting versus non-commuting coupling sectors to the bath's spectral weight.

At first sight, Eq. (58) looks alarming: because $\Sigma_+ \neq \Sigma_-$, the influence operator e^Δ is *not* manifestly Hermitian. The astute reader will, however, suspect that a non-Hermitian equilibrium state is a contradiction in terms, and they would be right.

The physically relevant quantity is the unnormalised reduced state

$$\bar{\rho}_Q(\beta) = e^{-\beta H_Q} e^{\Delta(\beta)}, \tag{81}$$

in which the bare Gibbs factor $e^{-\beta H_Q} = \mathrm{diag}(e^{-a}, e^{a})$ (with $a \equiv \beta\omega_q/2$) multiplies from the left. This thermal prefactor is precisely the agent of reconciliation. Setting

$$M \equiv \Delta_z\sigma_z + \Sigma_+\sigma_+ + \Sigma_-\sigma_-, \qquad \chi \equiv \sqrt{\Delta_z^2 + \Sigma^2}, \tag{82}$$

where $\Sigma^2 = \Sigma_+\Sigma_-$.

Since $M^2 = \chi^2\mathbb{I}$ (a pleasing consequence of the $\mathfrak{su}(2)$ algebra), the matrix exponential closes exactly:

$$e^\Delta = e^{\Delta_0}\left[\cosh\chi\,\mathbb{I} + \frac{\sinh\chi}{\chi}M\right]. \tag{83}$$

Factoring $e^{\Delta_0}\cosh\chi$ from e^Δ and defining $\hat{M} \equiv M/\chi$, the unnormalised state reads

$$\bar{\rho}_Q = e^{\Delta_0}\cosh\chi\, e^{-\beta H_Q}\left(\mathbb{I} + \tanh\chi\,\hat{M}\right), \tag{84}$$

Defining the scaled amplitude $\gamma \equiv \tanh\chi/\chi$, the normalised density matrix is

$$\rho_Q = \frac{1}{Z_Q}\begin{pmatrix} e^{-\beta\omega_q/2}(1+\gamma\Delta_z) & e^{-\beta\omega_q/2}\gamma\Sigma_+ \\ e^{\beta\omega_q/2}\gamma\Sigma_- & e^{\beta\omega_q/2}(1-\gamma\Delta_z)\end{pmatrix}. \tag{85}$$

with the normalisation

$$Z_Q = 2\left[\cosh\left(\frac{\beta\omega_q}{2}\right) - \gamma\Delta_z\sinh\left(\frac{\beta\omega_q}{2}\right)\right] \tag{86}$$

Now the KMS relation Eq. (76) does its work, and we see explicitly that the off-diagonal entries are real and equal. The apparent ladder asymmetry was a gauge artefact of writing the ladder decomposition before recombination with the bare system, and carried no physical content.

With the density matrix Eq. (85) in hand, the mean-force Hamiltonian $H_{\mathrm{MF}} = -\beta^{-1}\ln\rho_Q$ follows directly. Decomposing the state as $\rho_Q = \frac{1}{Z_Q}(A\,\mathbb{I} + h_z\sigma_z + h_x\sigma_x)$, with coefficients

$$h_z = \gamma\Delta_z\cosh\frac{\beta\omega_q}{2} - \sinh\frac{\beta\omega_q}{2}, \tag{87}$$

$$h_x = \gamma\Sigma, \tag{88}$$

where we used the KMS condition to identify $\Sigma = \sqrt{\Sigma_+\Sigma_-}$ in the transverse component. The identity coefficient A ensures unit trace but drops out of the Hamiltonian. The Bloch vector magnitude simplifies remarkably to $r = 2\chi/Z_Q$. Substituting this into the matrix logarithm, the normalisation factors Z_Q cancel, leaving the elegant compact form

$$H_{\mathrm{MF}} = \frac{2}{\beta\chi}\,\mathrm{arctanh}\left(\frac{2\chi}{Z_Q}\right)(h_z\sigma_z + h_x\sigma_x). \tag{89}$$

This expression constitutes the exact analytic solution of the problem. It is manifestly Hermitian (all parameters are real) and respects the equilibrium symmetries of the bath.

This little object contains a large amount of physics, but what is arguably most striking is the degree to which the effect of the bath has been compressed into just three numbers, $\mathcal{K}(0)$, $\mathcal{K}(\omega_q)$, and $\mathcal{R}(\omega_q)$.

NUMERICAL VALIDATION OF THE EXACT THEORY

Even in this simplest possible case, the physics of the reduced system is startlingly rich. The exact solution of the previous section therefore invites a detailed numerical analysis. With the exact form of the mean-force state, we can precisely characterise the behaviour of the system in different regimes of temperature and bath coupling.

First, to expose the coupling dependence cleanly, we reintroduce an explicit dimensionless strength g by writing the bath kernel as

$$K(u) = g^2 K_0(u). \tag{90}$$

All response kernels inherit the same quadratic scaling, such that

$$\Delta_z \to g^2\Delta_z, \qquad \Sigma_\pm \to g^2\Sigma_\pm, \qquad \chi \to g^2\chi, \tag{91}$$

The entire non-perturbative recombination of bare Hamiltonian and influence is therefore controlled by the single dimensionless dressing parameter χ through

$$\gamma(\chi) = \frac{\tanh\chi}{\chi}. \tag{92}$$

This apparently innocuous expression encodes the entire non-perturbative recombination of bare Hamiltonian and influence, and functions as an *order* parameter for the crossover between weak and ultrastrong coupling regimes. The reduced equilibrium state transitions between these regimes as χ passes through unity,

$$\chi(\beta, g, \theta) \sim 1 \qquad \Longleftrightarrow \qquad g^2 \sim \chi_0(\beta,\theta)^{-1}. \tag{93}$$

This crossover condition $\chi(\beta, g, \theta) = 1$ therefore defines a coupling-dependent temperature scale

$$g_\star(\beta) = \chi_0(\beta,\theta)^{-1/2}, \tag{94}$$

and an equivalent crossover temperature $\beta_\star(g)$. No approximation has been made; $g_\star$ is an exact property of the solution.

The function $\gamma(\chi)$ is elementary but worth dwelling on. It is monotone decreasing from $\gamma(0) = 1$ to $\gamma(\infty) = 0$, with curvature that changes sign near $\chi \approx 1.2$. In the two limiting regimes:

$$\chi \ll 1: \quad \gamma \simeq 1 - \tfrac{\chi^2}{3}, \tag{95}$$

$$\chi \gg 1: \quad \gamma \simeq \chi^{-1} = (g^2\chi_0)^{-1}. \tag{96}$$

These two expansions give a quantitative account of the two regimes identified qualitatively in Refs. [25, 26].

Weak-coupling regime, $\chi \ll 1$. When $\chi \ll 1$ (small g, or high temperature so that $\chi_0(\beta,\theta)$ is small), $\gamma \simeq 1$ and the density matrix Eq. (85) reduces to

$$\rho_Q \approx \frac{1}{\tilde{Z}} \begin{pmatrix} e^{-a}(1+\Delta_z) & e^{-a}\sqrt{\chi^2-\Delta_z^2} \\ e^{a}\sqrt{\chi^2-\Delta_z^2} & e^{a}(1-\Delta_z) \end{pmatrix} + O(\chi^3), \tag{97}$$

with $a = \beta\omega_q/2$. The off-diagonal entry—the bare-basis coherence—is $O(g^2)$, arising from $\Sigma_\pm \propto g^2$. The diagonal elements are similarly deformed from bare Gibbs at order g^2 through Δ_z. This is the perturbative dressing familiar from weak-coupling master equations, now derived exactly with no approximation on the spectral density.

Ultrastrong / low-temperature regime, $\chi \gg 1$. When $\chi \gg 1$ (large g, or very low temperature where χ_0 grows), $\tanh\chi \to 1$ and $\gamma \to \chi^{-1}$. The density matrix is then dominated by the normalised influence direction $\hat{M} = M/\chi$:

$$\rho_Q \approx \frac{e^{-\beta H_Q}\left(\mathbb{I} + \hat{M}\right)}{Z_Q}. \tag{98}$$

The state is sharply polarised along the axis defined by M, losing sensitivity to the bare Gibbs factor beyond what is needed to satisfy the KMS boundary conditions. The specific bath spectral weight matters only through the saturated invariants entering χ_0, not through any detailed frequency structure. In this limit the qubit effectively decouples from its own Hamiltonian and aligns with the interaction direction, the situation described as the "ultrastrong" mean-force fixed point in Refs. [25, 26].

The passage between these regimes is not a phase transition but a smooth crossover centered at $\chi \approx 1$. For the mixed coupling studied here, the crossover is marked by two distinct signatures in the system's differential response: a peak in the Bloch-angle susceptibility $\partial_g\varphi$ at $g_{\mathrm{peak},\varphi}$ and a peak in the Bloch-radius gradient $\partial_g r$ at $g_{\mathrm{peak},r}$. The location of the observable crossover signatures can be determined analytically by differentiating the exact Bloch-vector components with respect to g. Defining $a = \beta\omega_q/2$ and the dimensionless invariant $\chi = g^2\chi_0(\beta)$, the components m_x and m_z (fixing $\omega_q = 1$) are

$$m_x = \frac{\Sigma\tanh\chi}{\chi\cosh a - \Delta_z\sinh a\tanh\chi}, \tag{99}$$

$$m_z = \frac{\Delta_z\cosh a\tanh\chi - \chi\sinh a}{\chi\cosh a - \Delta_z\sinh a\tanh\chi}, \tag{100}$$

where $\Delta_z = g^2\Delta_{z0}$ and $\Sigma = g^2\Sigma_0$. The exact observable gradients are then given by

$$\partial_g\varphi = \frac{2g\Sigma_0\chi\sinh a\,\mathrm{sech}^2\chi}{\Sigma^2\tanh^2\chi + (\Delta_z\tanh\chi - \chi\tanh a)^2}, \tag{101}$$

$$\partial_g r = \frac{4g\chi_0\tanh\chi\,\mathrm{sech}^2\chi}{rZ_Q^2}. \tag{102}$$

The exact location of these crossover signatures can be determined by setting the derivatives of the gradients to zero. Since the g-dependence enters solely through the invariant $\chi = (g/g_\star)^2$, we express the gradients as functions of χ and seek their maxima. In the high-temperature

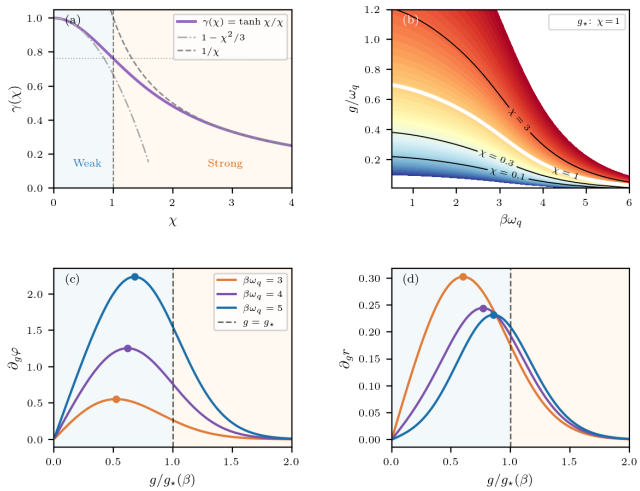

FIG. 1. Exact χ-crossover theory for the Ohmic bath (105) ($\theta = \pi/4$). *Panel (a)*: Universal crossover function $\gamma(\chi) = \tanh\chi/\chi$ between the weak-coupling ($1-\chi^2/3$) and ultrastrong ($1/\chi$) limits. *Panel (b)*: Map of $\chi(\beta, g) = g^2\chi_0(\beta)$ showing the low-T divergence of the crossover scale. The white curve marks the exact center $g_\star(\beta) = \chi_0(\beta)^{-1/2}$. *Panels (c,d)*: Observable signatures of the crossover in the Bloch-angle susceptibility $\partial_g\varphi$ and Bloch-radius gradient $\partial_g r$ plotted against the scaled coupling $g/g_\star(\beta)$. Rescaling the coupling aligns the peak positions for all temperatures at the same horizontal markers. Shaded regions demarcate the weak (blue) and strong (orange) coupling regimes, with filled circles indicating the analytical peak locations.

limit ($a \gg 1$), the condition $\frac{d}{dg}(\partial_g\varphi) = 0$ leads to the transcendental equation for the susceptibility peak:

$$1 - 4\chi\tanh\chi + \frac{4\chi c\,\mathrm{sech}^2\chi}{1 - c\tanh\chi} = 0, \tag{103}$$

where $c = \Delta_{z0}/\chi_0 \approx 0.7$ for the Ohmic bath at $\theta = \pi/4$. Solving this yields $\chi_{\mathrm{peak},\varphi} \approx 0.42$, pinning the peak at $g_{\mathrm{peak},\varphi} \approx 0.65\, g_\star(\beta)$. Similarly, the purity gradient $\partial_g r$ achieves its maximum when

$$1 + 2\chi\left[\coth\chi - 3\tanh\chi + \frac{2c\,\mathrm{sech}^2\chi}{1 - c\tanh\chi}\right] = 0, \tag{104}$$

which is satisfied at $\chi_{\mathrm{peak},r} \approx 0.85$, or $g_{\mathrm{peak},r} \approx 0.92\, g_\star(\beta)$. Both conditions confirm that the observable signatures are universal multiples of the bath response scale $g_\star(\beta) = \chi_0^{-1/2}$. Indeed, by rescaling the coupling as $\xi = g/g_\star(\beta)$, both crossover signatures are pinned to fixed universal values of the invariant χ, aligning the horizontal peak positions for all temperatures (Fig. 1c,d). This demonstrates that the mean-force crossover is fundamentally governed by the accumulated bath response scale regardless of the environment statistics.

To make these predictions concrete we commit to a specific bath that permits exact analytic evaluation of all quantities. We choose a windowed Ohmic spectral density

$$J(\omega) = g^2\,\omega\, e^{-\omega/\omega_c}, \qquad \omega \in [0, 2\omega_q], \tag{105}$$

with cutoff $\omega_c = 5\omega_q$. The UV window $[0, 2\omega_q]$ is chosen to match typical numerical simulations where the bath state is truncated, while the exponential cutoff regularises the high-frequency tail [59]. Having specified the bath, we further specialise simulations to $\theta = \pi/4$ and $\omega_q = 1$, as all other behaviour follows from the universal scaling. Figure 1 shows the exact χ-crossover theory for the Ohmic bath (105) ($\theta = \pi/4$). With these analytic predictions in hand, we now turn to comparison with numerical approximations.

NUMERICAL VALIDATION

The exact solution of Sec. depends on the bath only through the kernel evaluated at three frequencies, $\omega \in \{0, \pm\omega_q\}$. To test it we compare against Exact Diagonalisation (ED) of a truncated spin-boson model: the qubit is coupled to N_ω harmonic oscillators, each represented with a per-mode Fock cutoff $n_{\max}$. A key feature of this comparison is that the analytic kernel integrals $\mathcal{K}(\omega) = \int_0^\infty d\nu\, J(\nu)\,\mathcal{F}(\nu, \omega, \beta)$ can be evaluated in two ways: using the full continuum spectral density (the *continuous analytic* branch) or restricted to the same N_ω discrete modes seen by ED (the *discrete analytic* branch). Distinguishing between these is essential for identifying the true origin of numerical discrepancies in the strong-coupling regime.

Figure 2 shows the excited-state population p_{11} as a function of coupling strength and inverse temperature, each for $N_\omega = 40$ modes. At weak coupling ($\chi \ll 1$) all three curves coincide. However, as the system crosses into the strong-coupling regime ($\chi \gtrsim 1$), the analytic predictions separate from the ED results. The key to interpreting this gap lies in the branching between the analytic solution using either the *a continuous* (dotted orange line) or *discrete* (blue dashed line) spectral density. We see that as χ increases into the crossover region and beyond, the discrete analytic result decouples from the ED curve and instead converges back to the continuous spectrum analytic solution, correctly converging to the ultrastrong limit. This is a powerful result: it shows that the HMF theory is exact even for small, discrete mode sets, and that a discrete bath is physically capable of reaching the ultrastrong limit.

The fact that ED (green solid) remains significantly higher than its own infinite-cutoff analytic prediction is therefore not a failure of the theory, nor even a finite-mode artifact. Rather, it is a *representability bottleneck* in the dimension of those modes. In the ultrastrong limit, the coupled system seeks a displaced ground state (a polaron-like state) where each mode k is displaced from

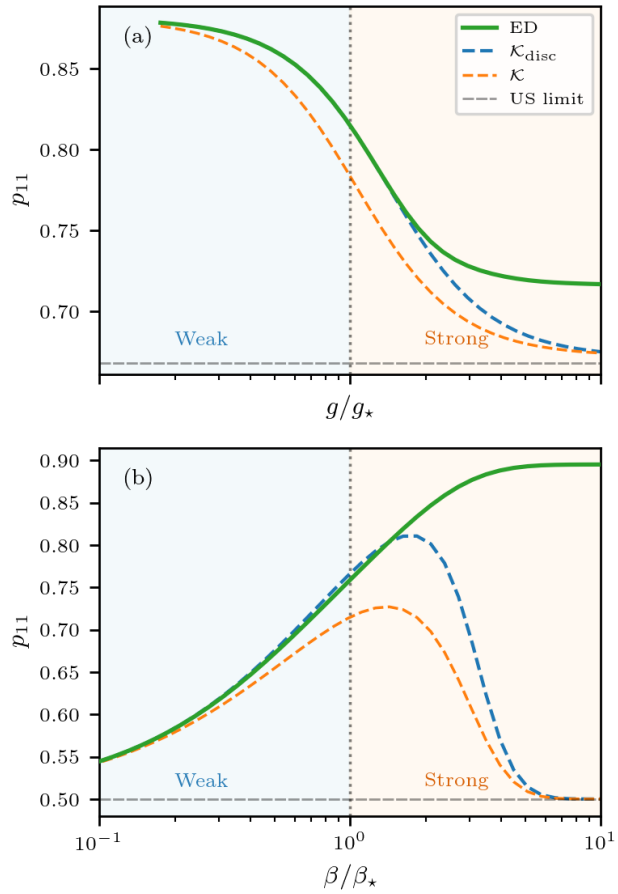

FIG. 2. Excited-state population p_{11} across the χ-crossover for $N_\omega = 40$ modes. Green solid: ED. Blue dashed: discrete analytic (same N_ω modes). Orange dotted: continuous analytic. Dashed horizontal lines mark the ultrastrong limit p_∞. The discrete analytic correctly reaches the US limit, matching the continuum. The gap left by ED (green) is a representability failure: the fixed Fock cutoff cannot accommodate the mode displacements required to reach the true HMF state.

vacuum by $\alpha_k \sim g_k/\omega_k$. Representing such a state faithfully requires a level cutoff $n_{\max} \gtrsim |\alpha_k|^2$. When $\chi \gtrsim 1$, the available Fock space in the ED simulation is insufficient to capture the tails of these displaced oscillators. The system is "trapped" in a subspace of artificially low occupation numbers, leading to over-estimated energies and the persistent gap observed at large χ.

The representability bottleneck implies that simulation accuracy should be highly sensitive to the choice of mode cutoffs as one enters the strong-coupling regime. Figure 3 explores this by monitoring the cutoff sensitivity Δp_{11} as a function of the spectral window bandwidth B (in units of ω_q).

The presence of a distinct peak in the sensitivity is a direct signature of the modes hitting the Fock boundary. As χ grows (lower temperatures), the requirement for mode displacements increases, pushing the available

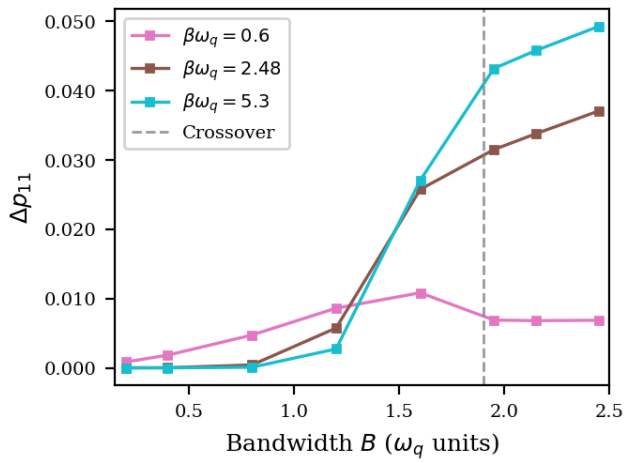

FIG. 3. Cutoff sensitivity $\Delta p_{11} \equiv |p_{11}(n_{\max} = 4) - p_{11}(n_{\max} = 6)|$ as a function of spectral bandwidth B (measured in units of ω_q). The vertical dashed line marks the crossover where mode occupation requirements begin to exceed the available Fock space. The sensitivity peaks sharply as the system enters the strong-coupling regime ($\chi \gtrsim 1$), marking the onset of the representability bottleneck. Higher sensitivity at lower temperatures reflects the increased mode-occupation required to represent the displaced ultrastrong state.

Fock space to its limit. This "simulability crossover" occurs exactly where $\chi \sim 1$, providing a concrete numerical marker for the validity of the finite-dimension approximation to the reduced equilibrium state.

In summary, the numerical results establish that the analytic HMF provides the correct physical target for both discrete and continuous baths. The deviations seen in simulation are not theoretical errors but markers of the intrinsic difficulty in representing the strongly-dressed open-system ground state within a finite Fock basis. The crossover parameter χ emerges as the universal controller of this simulability crossover, just as it governs the analytic structure of the state itself. It reveals that within the crossover regime, even the simplest systems contain rich nonlinearities in response to coupling and temperature, which cannot be efficiently accessed by standard numerical methods. More evidence - as if it were needed - of the fundamental importance of the exact strong coupling solution.

DISCUSSION

This paper has three distinct but interlocking contributions: a structural criterion for operator representability of the HMF, an exact closed-form solution in the transverse-field qubit model, and a numerical characterisation of the crossover between the weak- and ultrastrong-coupling regimes. We discuss each in turn

before drawing out the broader implications.

Closure and representability

The central algebraic result of Sec. is that the question "does a local closed-form HMF exist?" has a precise answer in terms of the operator algebra generated by repeated adjoint action of H_Q on the coupling f. Conditions (C1)–(C3) distil this into a hierarchy of closure requirements: adjoint closure of the coupling chain $\{f_n\}$, closure of that chain's quadratic span (which contains the Gaussian influence operator Δ), and BCH closure of commutators among the quadratic elements. When all three hold, $H_{\mathrm{MF}}(\beta)$ is a finite operator polynomial and can be extracted by a single BCH logarithm [48–50, 57]; when they fail, the HMF necessarily has support outside the target operator family and any representation within that family is a controlled approximation.

A notable consequence is the impossibility result for non-quadratic continuous-variable systems. For a Hamiltonian $H_Q = p^2/2m + V(x)$ with a polynomial potential $\deg V > 2$, the adjoint chain $\{f_n\}$ does not close in any finite-dimensional algebra: the chain grows without bound and generates the full universal enveloping algebra. This obstruction is analogous in spirit to the Groenewold–Van Hove theorem on consistent polynomial quantisation, and extends it to the equilibrium context: for such potentials no finite local operator can represent the exact reduced equilibrium state.

On the constructive side, the BCH form Eq. (53) provides a systematic generator of the HMF in any regime. The separation of operator structure (carried entirely by the adjoint chain) from all bath and temperature dependence (entering only through the scalar kernel moments $C^{>}_{\tilde{nm}}(\beta)$) means that, in practice, one need only compute these moments — themselves one-dimensional Laplace-type integrals over the spectral density — and then perform matrix algebra in the operator space. This is computationally tractable even when the adjoint chain is long, and suggests a natural numerical scheme: truncate the chain at a finite depth, compute the corresponding moments, and use the BCH formula to assemble the approximate HMF. The approximation is then purely on the operator side; all bath physics enters exactly.

The exact qubit solution

The spin-boson model is the canonical testbed for open-quantum-system physics, yet the exact HMF in the transverse-coupling case has not, to our knowledge, been written down explicitly outside of weak-coupling or ultrastrong limiting forms [25, 26, 60]. The result (89) achieves this using only three bath numbers: the kernel functional $\mathcal{K}$ evaluated at zero frequency, at the system resonance, and in the resonant channel $\mathcal{R}(\omega_q)$.

The compression of the entire bath into three scalars is exact, not perturbative, and follows directly from the $\mathfrak{su}(2)$ closure of the adjoint chain. The algebra guarantees that no higher-frequency evaluations can appear: all imaginary-time convolutions collapse, via the ordered Green's kernel $\mathcal{G}^{>}(x, y)$ and its closed-form reduction (68), to a finite set of Laplace transforms evaluated on the system's own mass shell. This is the imaginary-time analogue of the fact that, in linear response, a two-level system couples to the bath only at its transition frequency.

The crossover parameter $\chi = \sqrt{\Delta_z^2 + \Sigma^2}$ that emerges from the $\mathfrak{su}(2)$ algebra is more than a convenient shorthand. Its square is the total accumulated bath response seen by the qubit in imaginary time — the analogue of the polaron displacement energy — and $\gamma(\chi) = \tanh\chi/\chi$ is the universal renormalisation factor that interpolates between the two fixed points. At $\chi \ll 1$, $\gamma \approx 1$ and the state is a weakly dressed bare Gibbs distribution. At $\chi \gg 1$, $\gamma \sim 1/\chi$ and the state is pinned to the interaction direction, reproducing the ultrastrong fixed point of Cresser and Anders [25, 26] exactly. The crossover between these regimes is smooth, parameter-free, and analytically located at $\chi = 1$ — the precise condition $g = g_\star(\beta)$.

This solution also sheds light on an apparent paradox that arises in the intermediate steps of the calculation. The influence operator $\Delta(\beta)$ has manifestly asymmetric ladder amplitudes, $\Sigma_+ \neq \Sigma_-$, which would seem to imply a non-Hermitian reduced state. The resolution is that the asymmetry is a gauge artefact of computing the influence before recombining with the bare Gibbs factor $e^{-\beta H_Q}$: the KMS relation Eq. (76) precisely maps $\Sigma_+ = e^{\beta\omega_q}\Sigma_-$, and the product $e^{-\beta H_Q}e^{\Delta}$ is Hermitian by construction. The FDT is not an additional input but a consistency condition that is automatically enforced by the imaginary-time kernel.

The crossover as a representability bottleneck

The numerical results of Sec. add a dimension to the physical picture that is not visible from the analytic solution alone: the sensitivity of the simulated state to the representational limits of the Hilbert space.

The distinction between the discrete and continuous analytic branches in Fig. 2 is physically sharp. Both branches evaluate the same exact formula; crucially, the discrete-analytic branch (which uses the same mode set as ED) correctly tracks the continuum limit as $\chi \to \infty$. This proves that the HMF theory is exact for any mode set, and that a discrete bath possesses the same intrinsic ability to reach the ultrastrong limit as the continuum.

The divergence of ED from its own discrete analytic

prediction as $\chi \gtrsim 1$ therefore reveals a **representability bottleneck**. In the strongly coupled regime, the bath oscillators acquire large polaron-like displacements. Representing such highly occupied states in a simulation with a fixed Fock cutoff $n_{\max}$ is fundamentally impossible; the truncated Hilbert space cannot accommodate the tails of the displaced ground state, forcing the simulation to stay at an artificially high energy level. This is a "mode occupation" failure, not a discretization failure.

The bandwidth analysis of Fig. 3 makes this quantitative. Choice of bandwidth B determines how the displacement "load" needed to reach the US limit is distributed among the available modes. The sharp peak in cutoff sensitivity is a direct marker of the available modes hitting their Fock boundaries as they attempt to polarize. This is not an arbitrary convergence artifact; it is the physical signature of trying to represent a highly non-local equilibrium state (in terms of mode occupancy) within a restricted representational space. The crossover parameter χ governs both the analytic structure of the state and the boundary where this representability bottleneck becomes insurmountable.

Connections to the literature

This work sits at the intersection of several lines of inquiry.

The Cresser-Anders framework [25, 26] gives the leading-order weak-coupling and exact ultrastrong-coupling forms for the qubit mean-force state. The present result interpolates between these limits exactly, identifies χ as the relevant dimensionless variable throughout, and gives the crossover scale $g_\star(\beta)$ in closed form.

The Du-Chen finite-bath HMF [27] generalises the mean-force concept to finite reservoirs by introducing a pair of effective Hamiltonians for system and bath. Our discrete-analytic branch is the commuting Gaussian instance of this construction: replacing the continuum by a discrete mode set is precisely the finite-bath limit, and the parametric sensitivity we observe in Fig. 3 is a concrete numerical realisation of the bath-feedback effects they discuss analytically.

The Burke-Iles-Smith structural study [28] analyses the operator content of the HMF through correlation-function methods. The potential-renormalisation perspective [19] provides a complementary bath-induced operator characterisation. Our closure criterion (C1)–(C3) provides a further algebraic characterisation: operator growth in the HMF corresponds precisely to the failure of one or more closure conditions.

More broadly, methods that access $\bar{\rho}_Q(\beta)$ numerically — HEOM, stochastic Schrödinger/Liouville approaches [37, 40], chain mappings, and reaction-coordinate schemes [30–33] — are all, in a sense, performing the quenched-density average numerically. For a recent unifying overview connecting these approaches, see Ref. [39]. Our BCH/adjoint-chain organisation shows how to extract a closed-form operator representation from such an average whenever the algebra closes, and identifies the minimal set of bath evaluations required (three numbers in the qubit case). This suggests a hybrid programme: use numerical methods to compute kernel moments on a target spectral density, then assemble the HMF analytically via the BCH formula. Surveys of the growing thermodynamic strong-coupling programme can be found in Refs. [6, 7].

Limitations and outlook

The central analytical restriction of the present work is Gaussianity. The quenched representation and the cumulant expansion hold for any bath, but the exact collapse of Δ to a bilinear form in the adjoint chain relies on the bath cumulant hierarchy truncating at second order. Beyond the Gaussian sector, higher cumulants $K^{(n)}$ with $n \geq 3$ enter (10) and introduce products $f_{n_1} \cdots f_{n_k}$ of arbitrarily high order. Closure of such a family in a finite algebra is generically much harder. The non-Gaussian extension therefore represents the natural next step, and the adjoint-chain organisation already provides the skeleton: each additional cumulant order contributes a new layer of kernel moments multiplying operator monomials of corresponding degree, and the BCH structure persists.

Within the Gaussian sector, the qubit model studied here is the simplest closed algebra. The next natural target is multi-qubit systems, where the adjoint chain generated by H_Q will generically involve k-body Pauli strings. Closure then requires the target family to contain those strings, and the kernel-moment computation becomes a matrix problem in an enlarged operator space. Whether the HMF remains a two-body operator for a chain of qubits, or generates genuinely k-local terms, is a question our framework can in principle answer.

Finally, the crossover physics identified here has immediate consequences for the growing experimental literature on strong-coupling thermodynamics with superconducting circuits and semiconductor quantum dots [6, 7]. The crossover scale $g_\star(\beta) = \chi_0(\beta)^{-1/2}$ translates directly into a device-parameter threshold: below it, the bare Gibbs description is adequate; above it, the mean-force correction is non-perturbative. The quantitative agreement between the exact formula and ED simulation in the weak-coupling sector provides a parameter-free benchmark that can be used to validate strong-coupling simulation protocols in those settings.

Appendix A: Averaging of the Time-Ordered Gaussian Propagator

We provide the explicit derivation of the identity in Eq. (A12), which relates the expectation value of a time-ordered exponential driven by Gaussian noise to a bilocal influence phase. Consider the time-ordered propagator

$$W_\xi(\beta) = \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\, \xi(\tau)\tilde{f}(\tau)\right], \tag{A1}$$

where $\xi(\tau)$ is a centered Gaussian stochastic field with covariance $\langle \xi(\tau)\xi(\tau')\rangle = K(\tau-\tau')$, and $\tilde{f}(\tau)$ is a system operator. We wish to prove that

$$\begin{aligned}\langle W_\xi(\beta)\rangle_\xi = \exp\Bigg[\frac{1}{2}\int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau-\tau') \\ \times \mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right]\Bigg].\end{aligned} \tag{A2}$$

The Dyson expansion of the time-ordered exponential is given by

$$\begin{aligned}W_\xi(\beta) = \sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\int_0^\beta d\tau_1\cdots d\tau_n \\ \times \mathcal{T}_\tau\left[\xi(\tau_1)\tilde{f}(\tau_1)\cdots\xi(\tau_n)\tilde{f}(\tau_n)\right].\end{aligned} \tag{A3}$$

Because each $\xi(\tau_i)$ is a scalar (c-number), the time-ordering operator acts only on the operator factors. Thus, we may pull the stochastic fields out of the ordering:

$$\mathcal{T}_\tau\left[\xi_1\tilde{f}_1\cdots\xi_n\tilde{f}_n\right] = (\xi_1\cdots\xi_n)\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_n\right], \tag{A4}$$

where $\xi_i \equiv \xi(\tau_i)$ and $\tilde{f}_i \equiv \tilde{f}(\tau_i)$. Taking the stochastic average over ξ and noting that centered Gaussian noise has vanishing odd moments, the expansion becomes

$$\langle W_\xi\rangle = \sum_{p=0}^{\infty}\frac{1}{(2p)!}\int d^{2p}\tau\, \langle\xi_1\cdots\xi_{2p}\rangle\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_{2p}\right]. \tag{A5}$$

By Wick's theorem, the even moments are given by the sum over all pairings P:

$$\langle\xi_1\cdots\xi_{2p}\rangle = \sum_P\prod_{(i,j)\in P}K(\tau_i-\tau_j). \tag{A6}$$

We now consider the expansion of the claimed exponential result. Define the bilocal operator

$$X \equiv \frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau'\, K(\tau-\tau')\mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right]. \tag{A7}$$

The p-th term in the series for e^X is $(1/p!)X^p$. Introducing p independent pairs of integration variables (τ_{2r-1},τ_{2r}), we write

$$X^p = \left(\frac{1}{2}\right)^p\int d^{2p}\tau\prod_{r=1}^{p}\left[K_{(2r-1)(2r)}\mathcal{T}_\tau\left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right]\right], \tag{A8}$$

so that the full series is

$$\begin{aligned}e^X = \sum_{p=0}^{\infty}\frac{1}{p!}\left(\frac{1}{2}\right)^p\int d^{2p}\tau\left[\prod_{r=1}^{p}K_{(2r-1)(2r)}\right] \\ \times\left[\prod_{r=1}^{p}\mathcal{T}_\tau\left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right]\right].\end{aligned} \tag{A9}$$

To relate this to Eq. (A5), we exploit the property of time ordering on the hypercube $[0,\beta]^{2p}$. apart from a set of measure zero, the integration domain can be partitioned into $(2p)!$ disjoint regions R_π defined by the strict total ordering $\tau_{\pi(1)} > \tau_{\pi(2)} > \cdots > \tau_{\pi(2p)}$. On each such region, the global time ordering satisfies $\mathcal{T}_\tau[\tilde{f}_1\cdots\tilde{f}_{2p}] = \tilde{f}_{\pi(1)}\cdots\tilde{f}_{\pi(2p)}$. Similarly, each pairwise time-ordering $\mathcal{T}_\tau[\tilde{f}_{2r-1}\tilde{f}_{2r}]$ collapses to the correctly oriented product according to the same strict ordering. It follows that the product of pair-ordered operators coincides with the global ordering almost everywhere:

$$\prod_{r=1}^{p}\mathcal{T}_\tau\left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right] = \mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_{2p}\right]. \tag{A10}$$

Substituting this into Eq. (A9), we may pull the global time ordering out of the products.

The final step requires counting the occurrences of each Wick pairing. The expansion of X^p utilizes p labelled slots $r = 1,\ldots,p$, each containing an ordered pair. A Wick contraction P is a set of p unordered pairs. For any fixed Wick pairing P, there are $p!$ ways to assign the pairs to the p slots and 2^p ways to orient each pair. Each Wick pairing is therefore represented exactly $2^p p!$ times in the integrand. This multiplicity is canceled by the prefactor $(1/p!)(1/2)^p$, yielding

$$e^X = \sum_{p=0}^{\infty}\frac{1}{(2p)!}\int d^{2p}\tau\left(\sum_P\prod_{(i,j)\in P}K_{ij}\right)\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_{2p}\right]. \tag{A11}$$

This series matches Eq. (A5) term-by-term, proving the exponentiation of the Gaussian average. Consequently, we may right the influence on the reduced density matrix as

$$\begin{aligned}\langle W_\xi(\beta)\rangle_\xi = \exp\Bigg[\frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau' K(\tau-\tau') \\ \times\mathcal{T}_\tau[\tilde{f}(\tau)\tilde{f}(\tau')]\Bigg].\end{aligned} \tag{A12}$$

Domain Folding and the Triangular Form of the Influence Operator

The influence operator $\Delta(\beta)$

$$\Delta(\beta) = \frac{1}{2}\int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau-\tau')\, \mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big]. \quad \text{(A13)}$$

admits a simplified representation over a strictly ordered time domain. By definition, the time-ordering operator for two system operators acts as

$$\mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right] = \Theta(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau') + \Theta(\tau'-\tau)\tilde{f}(\tau')\tilde{f}(\tau). \quad \text{(A14)}$$

Substituting this into the integral and splitting the domain yields

$$\begin{aligned}\Delta(\beta) = \frac{1}{2}\iint_{\tau>\tau'} d\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau') \\ + \frac{1}{2}\iint_{\tau'>\tau} d\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau')\tilde{f}(\tau).\end{aligned} \quad \text{(A15)}$$

In the second term, we perform the swap of dummy variables $(\tau,\tau') \mapsto (\tau',\tau)$. The region $\tau' > \tau$ becomes $\tau > \tau'$, the operator product transforms to $\tilde{f}(\tau)\tilde{f}(\tau')$, and the kernel becomes $K(\tau'-\tau)$. Thus,

$$\begin{aligned}\frac{1}{2}\iint_{\tau'>\tau} d\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau')\tilde{f}(\tau) \\ = \frac{1}{2}\iint_{\tau>\tau'} d\tau d\tau'\, K(\tau'-\tau)\tilde{f}(\tau)\tilde{f}(\tau').\end{aligned} \quad \text{(A16)}$$

Using the symmetry of the equilibrium kernel, $K(\tau-\tau') = K(\tau'-\tau)$, we find that both contributions are identical on the ordered region. Summing them yields the triangular representation used throughout this work:

$$\Delta(\beta) = \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau'). \quad \text{(A17)}$$

Appendix B: Algebraic Construction in the Pauli Basis

In the main text (Sec.), the exact solution was derived using the ladder basis $\sigma_\pm$, which diagonalises the imaginary-time evolution and leads directly to the response kernels. Here we afford an alternative perspective by working directly in the Pauli basis $\{\sigma_x, \sigma_y, \sigma_z\}$. While algebraically heavier, this route makes the operator closure explicit at the level of structure constants and facilitates the systematic summation of the BCH series.

1. Polylogarithmic decomposition from the adjoint chain

We consider the same transverse coupling model defined by

$$H_Q = \frac{\omega_q}{2}\sigma_z, \qquad f = c\,\sigma_z - s\,\sigma_x, \quad \text{(B1)}$$

with $c = \cos\theta$, $s = \sin\theta$. The adjoint chain $f_n \equiv \mathrm{ad}^n_{H_Q}(f)$ is given by

$$f_n = \mathbf{v}_n \cdot \boldsymbol{\sigma}, \qquad \mathbf{v}_n = \begin{cases}(-s, 0, c) & n = 0, \\ (-s\omega_q^n, 0, 0) & n \geq 1,\ n \text{ even}, \\ (0, -is\omega_q^n, 0) & n \geq 1,\ n \text{ odd}.\end{cases} \quad \text{(B2)}$$

The recurrence ensures that $f_{n\geq 1}$ alternates between the x and y axes, while f_0 contains a persistent z-component.

The influence exponent $\Delta(\beta) = \sum_{n,m} C^{>}_{nm} f_n f_m$ is evaluated using the standard Pauli product rule $(\mathbf{a}\cdot\boldsymbol{\sigma})(\mathbf{b}\cdot\boldsymbol{\sigma}) = (\mathbf{a}\cdot\mathbf{b})\mathbb{I} + i(\mathbf{a}\times\mathbf{b})\cdot\boldsymbol{\sigma}$. Decomposition of the ordered moments into symmetric (S_{nm}) and antisymmetric (A_{nm}) parts reveals that only the commutator sector A_{nm} contributes to the operator structure:

$$\Delta(\beta) \cong 2i \sum_{n>m} A_{nm}\,(\mathbf{v}_n \times \mathbf{v}_m)\cdot\boldsymbol{\sigma}, \quad \text{(B3)}$$

up to a scalar shift $\Delta_0\mathbb{I}$. The cross products exhibit a strict parity selection rule:

- For $n, m \geq 1$: $\mathbf{v}_n \times \mathbf{v}_m$ lies along $\hat{\mathbf{z}}$ since both vectors are confined to the xy-plane.
- For $m = 0$: $\mathbf{v}_n \times \mathbf{v}_0$ generates terms in the xy-plane, mixing the longitudinal and transverse directions.

Explicitly summing these series yields the channel decomposition

$$\Delta = \Delta_0\mathbb{I} + \Delta_x\sigma_x + \Delta_y\sigma_y + \Delta_z\sigma_z, \quad \text{(B4)}$$

where the coefficients are polylogarithmic series in ω_q:

$$\Delta_x = -2cs \sum_{\ell\geq 0} A_{2\ell+1,0}(\beta)\,\omega_q^{2\ell+1}, \quad \text{(B5)}$$

$$\Delta_y = -2ics \sum_{\ell\geq 1} A_{2\ell,0}(\beta)\,\omega_q^{2\ell}, \quad \text{(B6)}$$

$$\begin{aligned}\Delta_z = -2s^2 \sum_{k\geq 1,\ell\geq 0} A_{2k,2\ell+1}(\beta)\,\omega_q^{2k+2\ell+1} \\ - 2s^2 \sum_{\ell\geq 0} A_{2\ell+1,0}(\beta)\,\omega_q^{2\ell+1}.\end{aligned} \quad \text{(B7)}$$

These series are the Pauli-basis equivalent of the closed-form expressions obtained in Sec. . The connection is established by identifying the sums as Taylor expansions of the kernels $\mathcal{G}^{>}(0, \pm\omega_q)$ and $\mathcal{G}^{>}(\pm\omega_q, \mp\omega_q)$.

2. Exact BCH resummation via the nilpotency identity

The central challenge in the mean-force construction is the evaluation of the BCH logarithm $-\beta H_{\rm MF} = \log(e^{-\beta H_Q} e^{\Delta})$. While the closure theorem guarantees that this remains within the algebra, a manual summation of the BCH series is laborious. Here we show how the $\mathfrak{su}(2)$ structure allows for an exact, non-perturbative resummation.

Let $M \equiv \Delta - \Delta_0 \mathbb{I} = \boldsymbol{\Delta} \cdot \boldsymbol{\sigma}$ be the traceless part of the influence operator. The square of this operator satisfies a crucial identity:

$$M^2 = (\boldsymbol{\Delta} \cdot \boldsymbol{\sigma})^2 = (\boldsymbol{\Delta} \cdot \boldsymbol{\Delta})\mathbb{I} \equiv \chi^2 \mathbb{I}, \tag{B8}$$

where $\chi^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$ (or $\Delta_z^2 + \Sigma_+\Sigma_-$ in the ladder gauge). This "generalized nilpotency" condition truncates the power series for the exponential e^M into even and odd sectors, yielding the exact Euler-like formula:

$$e^{\Delta} = e^{\Delta_0} \left[\cosh\chi\, \mathbb{I} + \frac{\sinh\chi}{\chi} M \right]. \tag{B9}$$

This expression sums the internal structure of the influence operator potential to all orders. The full reduced state is then the product of two $\mathrm{SL}(2,\mathbb{C})$ matrices:

$$\begin{aligned} \bar{\rho}_Q &= e^{-\beta H_Q} e^{\Delta} \\ &= e^{\Delta_0} \begin{pmatrix} e^{-a} & 0 \\ 0 & e^{a} \end{pmatrix} \begin{pmatrix} \cosh\chi + \varphi\Delta_z & \varphi(\Delta_x - i\Delta_y) \\ \varphi(\Delta_x + i\Delta_y) & \cosh\chi - \varphi\Delta_z \end{pmatrix} \end{aligned} \tag{B10}$$

with $a = \beta\omega_q/2$ and $\varphi = \sinh\chi/\chi$. Since the product of 2×2 matrices is a 2×2 matrix, the BCH series (which is the logarithm of this product) is strictly confined to the Pauli algebra.

Explicitly, for a resulting state $\rho_Q = \frac{1}{2}(\mathbb{I} + \mathbf{r}\cdot\boldsymbol{\sigma})$, the mean-force Hamiltonian is

$$H_{\rm MF} = c_0 \mathbb{I} - \frac{1}{\beta} \frac{\operatorname{arctanh} r}{r} \mathbf{r} \cdot \boldsymbol{\sigma}. \tag{B11}$$

This formula represents the analytic summation of the entire BCH commutator tower. The "closure" discussed in Sec. manifests here as the fact that χ is a finite scalar number, preventing the proliferation of independent operator directions. All nonlinearities in the coupling strength are packaged into the transcendental dependence of χ and $\mathbf{r}$ on the channel coefficients $\Delta_{x,y,z}$.

This work would have been impossible without the manifestations of linear algebra collectively termed AI. At every stage their assistance has been vital in navigating otherwise insuperable difficulties. They are however shiftless layabouts who will happily tell you the sky is blue, and are under no circumstances to be trusted.

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